15 The nth term of a number sequence is $n^2 + 7$

(a) Find the first three terms of this sequence.

$n = 1$: $1^2 + 7 = 8$

$n = 2$: $2^2 + 7 = 11$

$n = 3$: $3^2 + 7 = 16$

1 mark
Substitute 1, 2, 3

1 mark
8, 11, 16

(2)

128 is a term of this sequence.

(b) Which term?

$n^2 + 7 = 128$

$-7 \qquad -7$

$n^2 = 121$

$n = 11$, so it is the 11th term.

1 mark
11th term

(1)

(Total for Question 15 is 3 marks)